A Coarse-to-Fine Scheme for Unsupervised Nonlinear Hyperspectral Unmixing Based on an Extended Multilinear Mixing Model

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***Abstract*— Recently, the research on nonlinear unmixing for hyperspectral images (HSIs) has received more and more attention. However, unsupervised nonlinear unmixing methods that jointly estimate endmembers and abundances from HSIs are insufficiently studied. Besides, the reasonable description of the wavelength-dependent nonlinear intensity and the effective utilization of the spectral and spatial information of HSIs remain to be improved. Based on an extended multilinear mixing (EMLM) model, a coarse-to-fine scheme is proposed for unsupervised nonlinear hyperspectral unmixing to address the above issues. Coarse HSIs generated based on the superpixel segmentation are unmixed first, and then, fine unmixing on the original HSIs is achieved with the guidance of the coarse unmixing results. The endmembers extracted by the coarse unmixing are used to update the endmembers in the fine unmixing, and the coarse abundances and nonlinear parameters are integrated into regularizers. To be specific, a weighted sparse regularizer of abundances and a weighted graph regularizer of nonlinear parameters are constructed and incorporated into the objective function. In this way, some priors can be well modeled and exploited, including that the neighboring pixels share similar sparsity patterns in the abundances and show the consistency in correlations between different bands of the nonlinear parameters. Finally, the alternative optimization strategy (AOS) and the alternating direction method of multipliers (ADMM) are applied to derive the algorithm. Experimental results on the synthetic, laboratory-created, and real hyperspectral data demonstrate that the proposed method outperforms the state-of-the-art nonlinear unmixing methods.**

***Index Terms*— Hyperspectral images (HSIs), superpixel, unsupervised nonlinear unmixing.**

I. Introduction

HYPERSPECTRAL images (HSIs), captured by hyperspectral imaging spectrometers, record the spatial distribution of materials and narrow spectral bands covering the typical visible and near-infrared wavelengths. However, due to the limited spatial resolution, there are usually a large number of mixed pixels in HSIs. The main task of spectral unmixing is to decompose the mixed pixels to extract the spectra of various pure materials (i.e., endmembers) and determine their proportions (i.e., abundances) [1].

Based on the assumption that the light is scattered only once by one material before reaching the sensor, the linear mixture model (LMM) represents an observed pixel as a linear combination of endmembers. Due to its simple mathematical representation and clear physical meaning, the LMM has been widely used for unmixing. For example, the N-FINDR [2] and vertex component analysis (VCA) [3] are typical endmember extraction methods. With known endmembers, methods such as the fully constrained least squares (FCLS) [4] serve for the abundance estimation. Instead of adopting the strategy of estimating abundances with known endmembers (i.e., supervised unmixing), nonnegative matrix factorization (NMF) [5] jointly estimates the endmembers and abundances following an unsupervised mode. To address the nonconvexity of NMF and search for a good solution, various constraints are combined with the NMF-based unmixing framework, such as $\ell_{1/2}$ NMF (LNMF) [6], where the $\ell_{1/2}$ regularizer enforces the sparsity of abundances. However, in complicated areas, such as the intimate mixtures of microscopic mineral particles and the vegetation area with a multilayer structure, the light may interact with more than one material, which leads to significant nonlinear mixing effects [7]. Therefore, further research on nonlinear mixture models (NMMs) is urgently needed to properly solve the unmixing problem in real scenarios.

Over the past few decades, many NMM-based unmixing methods have emerged [8]. Radiative transfer models, such as the Hapke model [9], relate the reflectance to a set of real-world physical parameters to solve the nonlinear unmixing problem for intimate mixtures. However, in practical applications, the acquisition of intrinsic physical parameters is difficult. Therefore, to alleviate the overreliance on physical parameters, the bilinear mixture models (BMMs) further extend the LMM by introducing an additional bilinear term that models the second-order scatterings between materials. Typical BMMs include the Nascimento model [10], Fan model (FM) [11], generalized bilinear model (GBM) [12], and polynomial postnonlinear model (PPNM) [13]. Due to the simplicity of expressing nonlinearities, many BMM-based unmixing methods have been proposed, such as the abundance

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estimation algorithm by using the geometric characteristics of BMMs [14] and the BMM-based constrained NMF (BCNMF) [15]. However, the lack of consideration for higher order scatterings limits the application of BMMs in complex scenes.

The multilinear mixing (MLM) model [16] includes all orders of scatterings by introducing the scalar nonlinear parameters to describe the probability of light undergoing further interactions. In [17], an unsupervised nonlinear unmixing method based on the MLM is proposed, called MLMp, which estimates the endmembers jointly with the abundances and nonlinear parameters. With the simplified objective function in MLMp, the existing MLM-based unmixing methods achieve improved performance with regularizers, such as the $\ell_{2,1}$ norm [18] and graph regularizers [19]. Notably, these methods have an implicit assumption that all the bands share the same mixing mode. However, since each band of electromagnetic waves interacts with different ground covers in different ways, there may be varying degrees of multiple scatterings across the same pixel's bands [20], [21], [22].

To interpret the wavelength-dependent nonlinear intensity, an extended multilinear mixing (EMLM) model extends the MLM by vectorizing the scalar nonlinear parameters. Based on the EMLM model, a bandwise nonlinear spectral unmixing (BNLSU) method is proposed [23]. In BNLSU, both the abundances' sparsity and nonlinear parameters' smoothness are exploited to alleviate the local minima. In our previous work [24], a spectral–spatial reweighted robust nonlinear unmixing algorithm is proposed to improve the robustness and accuracy of the EMLM-based unmixing by using a robust metric and reweighted regularizers. Provided that the endmembers are known or pre-extracted, these methods only serve for the abundance estimation. However, the assumption of the existence of pure pixels in the HSIs often fails, resulting that the endmembers being extracted incorrectly, which further increases the abundance estimation error [25].

In recent years, remarkable progress has been made in deep learning-based unmixing methods [26]. For instance, an unsupervised spectral unmixing method utilizing a two-stream autoencoder network is designed in [27]. Taking advantage of a 3-D-CNN-based network, the researchers in [28] proposed an unsupervised nonlinear spectral unmixing method with additive nonlinear models. In [29], an unsupervised neural network-based scheme for nonlinear spectral unmixing is proposed, incorporating a kernel layer. Despite the success achieved by deep learning techniques in unmixing tasks, certain limitations still persist, e.g., the lack of physical interpretability and the reliance on extensive training datasets. Consequently, our focus in this article remains on investigating model-driven unmixing methods.

Meanwhile, to achieve high-accuracy unmixing, it is crucial to integrate priors into unmixing problems, particularly nonlinear unmixing ones [30]. Accurate modeling and effective utilization of priors can narrow down the solution space, leading to better results. Instead of being evenly distributed across the scene, homogeneous materials often appear in local regions of HSIs. Besides, neighboring pixels exhibit strong spatial correlations. Numerous studies have been devoted to designing regularizers to reasonably model such priors. The well-known total variation regularizer, exploiting the spatial information at a level of the first-order pixel neighborhood, is used in sparse unmixing to enforce the piecewise smoothness of abundances [31]. To leverage the high-order neighborhood information, several studies have focused on HSI processing with superpixels [32]. A superpixel-based reweighted low-rank and total variation sparse unmixing method [33] and a kernel-based nonlinear unmixing method using superpixels [34] were proposed to consider the spatial proximity and the spectral similarity. Moreover, a recent study [35] introduces a parallel implementation of the graph-based spectral unmixing with superpixel structures. Usually, spatial regularizers are combined with sparse constraints rather than being imposed solely. However, due to the overreliance on parameter settings, more efforts are required to tune the spatial regularizers' parameters. Besides, the significant increase in the complexity of the unmixing problem results in a heavy computational burden [36].

To circumvent the above difficulties, a recent trend is to integrate the spatially informative weights into the sparse regularizers. In [37], a spectral–spatial weighting factor is incorporated into sparse unmixing with the ℓ_1 norm-based regularizer, which exploits both the spectral and spatial information in HSIs. Combining the spatial prior with the sparsity of abundances, a spatial group sparsity regularizer is constructed and integrated into the NMF-based unmixing framework [38]. In [39], a precalculated spatial weighting factor is introduced to preserve the original spatial property of HSIs. However, these well-designed regularizers are only used for linear unmixing, limiting their potential. On the other hand, although the "coarse-to-fine" idea is implicit in the unmixing process of the above methods, there is no research to generalize and further combine it with the nonlinear unmixing problem.

To this end, the motivation of this article is to propose an unsupervised nonlinear unmixing method that can not only account for the wavelength-dependent nonlinear intensity but also can achieve high-accuracy estimates of both the abundances and endmembers. The current research on the EMLM is still at a preliminary stage, which motivates us to conduct more in-depth investigations. Although unsupervised unmixing to jointly estimate the endmembers and abundances has wide applications in real-world scenarios, the appropriate construction of an unsupervised unmixing framework and effective optimization algorithms to solve the related problems remains a great challenge. Moreover, for unmixing problems, especially the nonlinear unmixing problems with more free parameters, designing regularizers to accurately model the priors that well exploit the rich spectral and spatial information in HSIs is crucial for improving unmixing accuracy.

To solve the above problems, based on the EMLM model, a coarse-to-fine scheme for unsupervised nonlinear hyperspectral unmixing, named CTF-UNHU, is proposed. First, superpixel segmentation is performed on the original HSIs as a preprocessing step. Then, the coarse HSIs constructed by averaging the pixels' spectra at each superpixel are unmixed, referred to as coarse unmixing, the results of which are

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leveraged to guide the fine unmixing on the original HSIs. Specifically, the endmembers extracted by the coarse unmixing are used to update the endmembers in the fine unmixing, and the coarse abundances and nonlinear parameters are integrated into the regularizer design. It can be considered that the neighboring pixels commonly share similar sparsity patterns in the abundances and show the consistency in correlations between the nonlinear parameters at different bands. These priors can be well modeled by a weighted sparse regularizer of abundances and a weighted graph regularizer of nonlinear parameters, respectively. Employing the two regularizers, a constrained optimization problem to achieve fine unmixing is formulated. Finally, the algorithm is derived based on the alternative optimization strategy (AOS) [40] and the alternating direction method of multipliers (ADMM) [41].

The main contributions of this article can be briefly summarized as follows:

1) An unsupervised nonlinear unmixing algorithm is proposed based on the EMLM model, which can not only describe the wavelength-dependent nonlinear intensity but also achieve the joint estimation of the endmembers and abundances. Moreover, a coarse-to-fine unmixing scheme is designed, which provides a new idea to express the spatial information of HSIs during the unmixing process.
2) Considering the spectral and spatial priors for the abundances and nonlinear parameters by leveraging the coarse unmixing results, a weighted sparse regularizer of abundances and a weighted graph regularizer of nonlinear parameters are designed and incorporated into the objective function to overcome overfitting and improve unmixing accuracy.
3) The AOS and ADMM are used for the optimization of the unmixing problem. Extensive ablation experiments and comparative experiments are conducted to analyze the impact of the novelty of the proposed method on the unmixing accuracy and to validate the superiority of the proposed method over the state-of-the-art methods.

The remainder of this article is organized as follows. Section I briefly reviews the LMM, MLM, and EMLM. Next, the formulations of the constrained optimization problems and the detailed derivation of the proposed method are provided in Section II. In Section III, experiments on the synthetic, laboratory-created, and real hyperspectral data are conducted. Finally, Section IV gives the conclusion of this article.

II. Spectral Mixture Models

A. LMM

Given the matrix of HSIs $\boldsymbol{X} \in \mathbb{R}^{n\times m}$ containing n bands and m pixels. Based on the assumption of the LMM, each pixel $\boldsymbol{x}_j \in \mathbb{R}^n$ $(j = 1, \ldots, m)$ is modeled by a linear combination of the endmembers according to their abundances. The abundances should satisfy two constraints, i.e., the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC). The LMM can be mathematically denoted as

$$\begin{aligned} \boldsymbol{x}_j &= \sum_{i=1}^{r} \boldsymbol{e}_i a_{i,j} + \boldsymbol{\varepsilon}_j \\ \text{s.t. } a_{i,j} &\geq 0, \quad \sum_{i=1}^{r} a_{i,j} = 1 \end{aligned} \tag{1}$$

where $\boldsymbol{e}_i \in \mathbb{R}^n$ is the ith endmember in the endmember matrix $\boldsymbol{E} \in \mathbb{R}^{n\times r}$, r is the number of the endmembers, $a_{i,j}$ is the ith endmember's abundance in $\boldsymbol{x}_j$, and $\boldsymbol{\varepsilon}_j \in \mathbb{R}^n$ is noise.

B. MLM and EMLM

The MLM model [16] further extends the BMMs to include all orders of interactions between endmembers. It is derived based on several assumptions. For example, before reaching the sensor, the light is assumed to interact with at least one material, and for the jth pixel, the probabilities of further interactions are denoted by p_j and $(1 - p_j)$, respectively. Besides, the probability of the light interacting with one material is proportional to its abundance, and the intensity of the light scattered by one material is related to that material's reflectance. Therefore, the formulation of the MLM model is written as

$$\begin{aligned} \boldsymbol{x}_j &= \left(1 - p_j\right) \sum_{i=1}^{r} \boldsymbol{e}_i a_{i,j} + \left(1 - p_j\right) p_j \sum_{i,k=1}^{r} (\boldsymbol{e}_i \odot \boldsymbol{e}_k) a_{i,j} a_{k,j} \\ &\quad + \left(1 - p_j\right) p_j^2 \sum_{i,k,l=1}^{r} (\boldsymbol{e}_i \odot \boldsymbol{e}_k \odot \boldsymbol{e}_l) a_{i,j} a_{k,j} a_{l,j} + \ldots + \boldsymbol{\varepsilon}_j \\ &= \left(1 - p_j\right) \boldsymbol{y}_j + p_j \boldsymbol{y}_j \odot \boldsymbol{x}_j + \boldsymbol{\varepsilon}_j \end{aligned} \tag{2}$$

where $\boldsymbol{y}_j = \sum_{i=1}^{r} \boldsymbol{e}_i a_{i,j} = \boldsymbol{E}\boldsymbol{a}_j$ is the linear component, and the ANC and ASC are also imposed. The pixel-dependent scalar nonlinear parameter p_j $(j = 1, \ldots, m)$ should be less than 1.

Notably, the above models are all based on an implicit assumption that all the bands in HSIs share the same mixing mode. However, this assumption may not well match the real scenes, where the mixing mode can be complex [22]. To address the major flaw in that assumption, the EMLM model [23] vectorizes the scalar nonlinear parameters of the MLM model to interpret the wavelength-dependent nonlinear intensity, which can be derived as

$$\boldsymbol{x}_j = \left(\boldsymbol{1}_n - \boldsymbol{P}_j\right) \odot \boldsymbol{y}_j + \boldsymbol{P}_j \odot \boldsymbol{y}_j \odot \boldsymbol{x}_j + \boldsymbol{\varepsilon}_j \tag{3}$$

where $\boldsymbol{1}_n \in \mathbb{R}^n$ is a vector of all 1 s, and $\boldsymbol{P}_j \in \mathbb{R}^n$ $(j = 1, \ldots, m)$ is the vectorized nonlinear parameter of the jth pixel. Similarly, the ANC and ASC are satisfied and a boundary constraint should be considered, i.e., $\boldsymbol{P}_j < \boldsymbol{1}_n$. It is noted that $\boldsymbol{P}_j$ can be negative, considering that a pixel may receive additional illumination from 3-D structures outside its instantaneous field of view. More detailed illustrations can be found in literatures [16], [23].

III. Proposed Method

The flowchart in Fig. 1 illustrates the proposed method. Superpixel segmentation is applied to the original HSIs $\boldsymbol{X}$,

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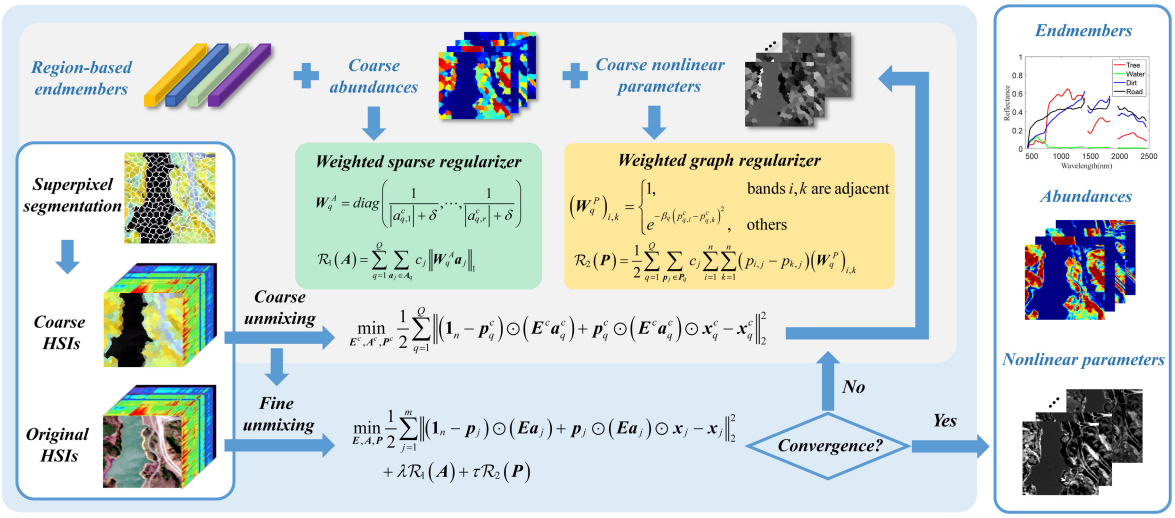


Fig. 1. Flowchart of CTF-UNHU.

followed by reordering the pixels in $\boldsymbol{X}$ based on the superpixel labels. The coarse HSIs $\boldsymbol{X}^c \in \mathbb{R}^{n\times Q}$ are generated by substituting the pixels' spectra with their averaged spectrum in each superpixel, where Q is the number of superpixels. $\boldsymbol{X}^c$ is unmixed to yield the region-based endmembers, $\boldsymbol{E}^c \in \mathbb{R}^{n\times r}$, coarse abundances, and nonlinear parameters, $\boldsymbol{A}^c \in \mathbb{R}^{r\times Q}$ and $\boldsymbol{P}^c \in \mathbb{R}^{n\times Q}$. The unmixing process of $\boldsymbol{X}$ and $\boldsymbol{X}^c$ is referred to as "fine unmixing" and "coarse unmixing" for brevity.

The coarse unmixing is integrated into the fine unmixing process, rather than being performed independently. During each iteration of the fine unmixing, the coarse unmixing is executed, and the outputs $\boldsymbol{E}^c$, $\boldsymbol{A}^c$, and $\boldsymbol{P}^c$ are individually utilized to update the endmembers, and the weight matrices $\boldsymbol{W}_q^A \in \mathbb{R}^{r\times r}$ and $\boldsymbol{W}_q^P \in \mathbb{R}^{n\times n}$ $(q = 1, \ldots, Q)$ in $\mathcal{R}_1(\boldsymbol{A})$ and $\mathcal{R}_2(\boldsymbol{P})$. After the two regularizers are incorporated into the EMLM-based objective function, the constrained optimization problem is determined and can be solved using the ADMM.

The fine unmixing proceeds to the next iteration when the convergence criteria is not satisfied. At this point, the endmembers and the superpixel-based weighted average of abundances and nonlinear parameters, updated in the previous iteration of the fine unmixing, are used to initialize $\boldsymbol{E}^c$, $\boldsymbol{A}^c$, and $\boldsymbol{P}^c$, respectively. Therefore, the fine unmixing outputs exert further positive influence on the coarse unmixing. The fine unmixing and coarse unmixing mutually reinforce each other, and after a certain number of iterations, the algorithm achieves convergence, providing accurate estimates of endmembers, abundances, and nonlinear parameters.

A. Coarse Unmixing

To define the local spatial neighborhood for each pixel, we employ the simple linear iterative clustering (SLIC) [32] to perform superpixel segmentation on the original HSIs $\boldsymbol{X}$. A coarse HSIs matrix $\boldsymbol{X}^c$ can be constructed with each column corresponding to the mean spectrum of all pixels within a superpixel. The EMLM-based constraint optimization problem for unmixing $\boldsymbol{X}^c$ can be formulated as follows:

$$\begin{aligned}
&\min_{E^c, A^c, P^c} L\left(\boldsymbol{E}^c, \boldsymbol{A}^c, \boldsymbol{P}^c\right) \\
&\text{s.t. } L\left(\boldsymbol{E}^c, \boldsymbol{A}^c, \boldsymbol{P}^c\right) \\
&\quad = \frac{1}{2}\sum_{q=1}^{Q}\left\|\left(\mathbf{1}_n - \boldsymbol{p}_q^c\right)\odot \boldsymbol{y}_q^c + \boldsymbol{p}_q^c \odot \boldsymbol{y}_q^c \odot \boldsymbol{x}_q^c - \boldsymbol{x}_q^c\right\|_2^2 \\
&\quad \boldsymbol{y}_q^c = \boldsymbol{E}^c\boldsymbol{a}_q^c, \boldsymbol{E}^c \geq 0, \boldsymbol{a}_q^c \geq 0, \mathbf{1}_r^\top \boldsymbol{a}_q^c = 1, \boldsymbol{p}_q^c < \mathbf{1}_n \quad (4)
\end{aligned}$$

where $\boldsymbol{x}_q^c$, $\boldsymbol{a}_q^c$, and $\boldsymbol{p}_q^c$ correspond to the components of $\boldsymbol{X}^c$, $\boldsymbol{A}^c$, and $\boldsymbol{P}^c$ in the qth superpixel, respectively. The abundances meet the ANC and ASC, and the nonlinear parameters should meet the boundary constraint. Moreover, the nonnegative constraint of endmembers is considered [17].

An efficient algorithm based on the AOS [42], i.e., iteratively optimizing each variable with the others fixed, is derived to solve the problem in (4), The pseudocode is provided to help understand the coarse unmixing process, which follows a similar derivation process as presented in Section III-C. Once stopping conditions are met, the coarse unmixing results are obtained, including the region-based endmembers $\boldsymbol{E}^c$, coarse abundances $\boldsymbol{A}^c$, and coarse nonlinear parameters $\boldsymbol{P}^c$.

It is noted that although a simple linear unmixing method such as the FCLS for the coarse unmixing may reduce the computational burden, it is imperative to employ Algorithm 1 for the coarse unmixing to address the nonlinear mixing effects, impose constraints on nonlinear parameters, and achieve accurate endmember extraction.

B. Regularizers Design Using Coarse Unmixing Results

The integration of HSIs' spectral and spatial information has been shown to improve unmixing performance in previous studies [24], [37]. Incorporating priors is crucial for solving the EMLM-based unmixing problem, as it helps narrow the scope of the solution space, improve the search for the optimal solution, and alleviate the influence of noise. In the proposed method, the priors capturing the local spatial similarity of

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Algorithm 1 Coarse Unmixing

Input: Coarse HSIs $\boldsymbol{X}^c \in \mathbb{R}^{n\times Q}$.
Output: Region-based endmembers $\boldsymbol{E}^c \in \mathbb{R}^{n\times r}$, coarse abundances $\boldsymbol{A}^c \in \mathbb{R}^{r\times Q}$, and coarse nonlinear parameters $\boldsymbol{P}^c \in \mathbb{R}^{n\times Q}$.

Set $i = 0$.
Initialization: $\boldsymbol{E}^{c(0)}$, $\boldsymbol{A}^{c(0)}$, and $\boldsymbol{P}^{c(0)}$.
While stopping conditions are not met, **do**
Update $\boldsymbol{E}^c$, $\min_{\boldsymbol{E}^c} L\left(\boldsymbol{E}^c, \boldsymbol{A}^{c(i)}, \boldsymbol{P}^{c(i)}\right) \quad s.t. \boldsymbol{E}^c \geq 0.$
Update $\boldsymbol{A}^c$, $\min_{\boldsymbol{A}^c} L\left(\boldsymbol{E}^{c(i+1)}, \boldsymbol{A}^c, \boldsymbol{P}^{c(i)}\right) \quad s.t. \boldsymbol{a}_q^c \geq 0, \mathbf{1}_r^\top \boldsymbol{a}_q^c = 1.$
Update $\boldsymbol{P}^c$, $\min_{\boldsymbol{P}^c} L\left(\boldsymbol{E}^{c(i+1)}, \boldsymbol{A}^{c(i+1)}, \boldsymbol{P}^c\right) \quad s.t. \boldsymbol{p}_q^c < \mathbf{1}_n.$
$i = i + 1$.
End

HSIs are incorporated into the unmixing process, guided by the coarse unmixing results. Moreover, the properties of abundances and nonlinear parameters in both spectral and spatial dimensions are comprehensively considered, and it is expected to incorporate more priors with fewer regularizers. Leveraging the coarse abundances and nonlinear parameters, the spatially informative weight matrices are constructed and integrated into regularizers to describe the spectral-dimensional nature of abundances and nonlinear parameters at the spatial neighborhood level. As a result, two regularizers are designed for abundances and nonlinear parameters, respectively.

1) Weighted Sparse Regularizer: Due to the local spatial distribution of ground covers, endmembers are less likely to occur in all the pixels. In other words, the abundances of most endmembers in a pixel may be equal to zero, known as the sparsity of abundances. Furthermore, since the proportions of different components can be locally similar, the neighboring pixels are expected to share the same sparsity pattern in the abundance matrix. The aforementioned physical priors can be mathematically represented with a weighted sparse regularizer of abundances.

$$\mathcal{R}_1(\boldsymbol{A}) = \sum_{q=1}^{Q} \sum_{\boldsymbol{a}_j \in \boldsymbol{A}_q} c_j \left\| \boldsymbol{W}_q^A \boldsymbol{a}_j \right\|_1. \tag{5}$$

According to superpixel labels, the abundance matrix $\boldsymbol{A}$ is first reordered as $\boldsymbol{A} = [\boldsymbol{A}_1, \ldots, \boldsymbol{A}_q, \ldots, \boldsymbol{A}_Q]$, where $\boldsymbol{A}_q \in \mathbb{R}^{r\times N_q}$ is the abundance submatrix corresponding to the qth superpixel and N_q is the number of pixels within the qth superpixel. Then, the spatially informative weight matrix $\boldsymbol{W}_q^A$ is integrated into the ℓ_1-norm-guided sparse regularizer, and $\boldsymbol{A}_q$ preserves similar sparsity patterns. Leveraging the estimated coarse abundances $\boldsymbol{a}_q^c$, the diagonal matrix $\boldsymbol{W}_q^A$ is constructed as

$$\boldsymbol{W}_q^A = \operatorname{diag}\left(\frac{1}{\left|a_{q,1}^c\right| + \delta}, \ldots, \frac{1}{\left|a_{q,r}^c\right| + \delta} \right) \tag{6}$$

where $\boldsymbol{a}_q^c = [a_{q,1}^c, \ldots, a_{q,r}^c]^\top$ and δ is a small positive value, which guarantees that the denominator is nonzero.

Then, the superpixelwise weight $\boldsymbol{W}_q^A$ is rescaled with the pixelwise defined factor c_j. c_j is inversely proportional to d_j, where d_j is the spectral–spatial distance between the jth pixel $\boldsymbol{x}_j$ and the cluster center $\boldsymbol{C}_q$ of the superpixel where it is located. Predefined in the segmentation process, $\boldsymbol{C}_q = [(\boldsymbol{x}_q^c)^\top, m_q, n_q]^\top \in \mathbb{R}^{n+2}$, where (m_q, n_q) is the spatial coordinate of $\boldsymbol{C}_q$. By combining the spectral angle distance $d_{spe} = \cos^{-1}(\boldsymbol{x}_j^\top \boldsymbol{x}_q^c / (||\boldsymbol{x}_j||_2 ||\boldsymbol{x}_q^c||_2))$ and the spatial Euclidean distance $d_{spa} = ((m_j - m_q)^2 + (n_j - n_q)^2)^{1/2}$, d_j is calculated as

$$d_j = \sqrt{d_{spe}^2 + \left(d_{spa}/\kappa\right)^2 \kappa_s^2} \tag{7}$$

where (m_j, n_j) is the spatial coordinate of $\boldsymbol{x}_j$. κ and κ_s are two parameters that control the average size of the superpixels and balance the relative importance between the spectral and spatial similarity, respectively.

In (5), it is clear that as the factor c_j increases, $\left\| \boldsymbol{W}_q^A \boldsymbol{a}_j \right\|_1$ turns to be small and vice versa. Rescaling with the factor c_j effectively prevents the loss of spatial details within superpixels. This is particularly important for small objects and boundary pixels that may not adhere to the designed nonzero patterns. Small objects often exhibit sparse structures that do not align with the sparsity structures of neighboring pixels. Boundary pixels are likely to be mixed pixels with dense abundances. Due to the spectral or spatial disparities in such heterogeneous pixels, the corresponding factor c_j can be relatively small. Consequently, incorporating the factor c_j to some extent mitigates boundary effects and prevents information loss associated with small objects.

2) Weighted Graph Regularizer: In real scenarios, electromagnetic waves of each wavelength interact with different ground covers in different ways, leading to differences in nonlinear contributions between spectral bands. The prior that the nonlinear intensities are similar at adjacent bands has been confirmed and applied in recent studies [23], [24]. Moreover, the similarity of nonlinear intensities among locally neighboring pixels should also be considered. These priors can be mathematically represented with a weighted graph regularizer

$$\begin{aligned} \mathcal{R}_2(\boldsymbol{P}) &= \frac{1}{2} \sum_{q=1}^{Q} \sum_{\boldsymbol{p}_j \in \boldsymbol{P}_q} c_j \sum_{i=1}^{n} \sum_{k=1}^{n} \left(p_{i,j} - p_{k,j}\right) \left(\boldsymbol{W}_q^P\right)_{i,k} \\ &= \frac{1}{2} \sum_{q=1}^{Q} \sum_{\boldsymbol{p}_j \in \boldsymbol{P}_q} c_j \boldsymbol{p}_j^\top \boldsymbol{L}_q \boldsymbol{p}_j \end{aligned} \tag{8}$$

where $\boldsymbol{L}_q = \boldsymbol{D}_q - \boldsymbol{W}_q^P$ is the graph Laplacian matrix, and $\boldsymbol{D}_q$ is a diagonal matrix with $(\boldsymbol{D}_q)_{i,i} = \sum_{k=1}^{n} (\boldsymbol{W}_q^P)_{i,k}$ [43].

Similarly, the nonlinear parameter matrix is reordered as $\boldsymbol{P} = [\boldsymbol{P}_1, \ldots, \boldsymbol{P}_q, \ldots, \boldsymbol{P}_Q]$, where $\boldsymbol{P}_q \in \mathbb{R}^{n\times N_q}$ is the nonlinear parameter submatrix of $\boldsymbol{P}$ for the qth superpixel. To explore the band-to-band correlation of the nonlinear parameters and preserve that in neighboring pixels, the estimated coarse nonlinear parameters $\boldsymbol{p}_q^c$ are mapped into a graph $\boldsymbol{G}_q$ of which each node represents a scalar nonlinear parameter at a certain band. The weight matrix $\boldsymbol{W}_q^P$ related to $\boldsymbol{G}_q$ is defined by

$$\left(\boldsymbol{W}_q^P\right)_{i,k} = \begin{cases} 1, & \text{bands } i, k \text{ are adjacent} \\ e^{-\beta_q \left(p_{q,i}^c - p_{q,k}^c\right)^2}, & \text{others} \end{cases} \tag{9}$$

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where $(\boldsymbol{W}_q^P)_{i,k}$ is an element in the ith row and the kth column of $\boldsymbol{W}_q^P$, $\boldsymbol{p}_q^c = [p_{q,1}^c, \ldots, p_{q,n}^c]^\top$, and the parameter β_q can be calculated using an empirical formula referring to [29]

$$\beta_q = \left(\frac{1}{n^2} \sum_{i=1}^{n} \sum_{k=1}^{n} \left(p_{q,i}^c - p_{q,k}^c \right)^2 \right)^{-2}. \tag{10}$$

The weight matrix $\boldsymbol{W}_q^P$ stores the weights corresponding to the edges connecting the nodes in the graph $\boldsymbol{G}_q$, which reflects the correlations of nonlinear parameters at adjacent/nonadjacent bands. Moreover, different degrees of constraint can be imposed by using the coefficient c_j.

C. Optimization Using the ADMM to Achieve Fine Unmixing

After incorporating $\mathcal{R}_1(\boldsymbol{A})$ and $\mathcal{R}_2(\boldsymbol{P})$ into the objective function, the optimization problem is formulated as

$$\begin{aligned} \min_{\boldsymbol{E},\boldsymbol{A},\boldsymbol{P}} \ & \frac{1}{2} \sum_{j=1}^{m} \left\| \left(\mathbf{1}_n - \boldsymbol{p}_j \right) \odot \boldsymbol{y}_j + \boldsymbol{p}_j \odot \boldsymbol{y}_j \odot \boldsymbol{x}_j - \boldsymbol{x}_j \right\|_2^2 \\ & + \lambda \mathcal{R}_1(\boldsymbol{A}) + \tau \mathcal{R}_2(\boldsymbol{P}) \\ \text{s.t.} \ & \boldsymbol{y}_j = \boldsymbol{E}\boldsymbol{a}_j, \boldsymbol{E} \geq 0, \boldsymbol{a}_j \geq 0, \mathbf{1}_r^\top \boldsymbol{a}_j = 1, \boldsymbol{p}_j < \mathbf{1}_n \end{aligned} \tag{11}$$

where λ and τ are two parameters to balance the contributions of the two regularizers $\mathcal{R}_1(\boldsymbol{A})$ and $\mathcal{R}_2(\boldsymbol{P})$.

The well-known ADMM is employed to solve the above problem. Decomposing the primal variables into several new variables corresponding to the optimization problem items, ADMM transforms the original complex optimization problem into a set of simple subproblems and effectively searches for the optimal solution by alternately updating all variables. After introducing two auxiliary variables $\boldsymbol{V}_1$ and $\boldsymbol{V}_2$, the problem can be reformulated as

$$\begin{aligned} \arg\min_{\boldsymbol{E},\boldsymbol{A},\boldsymbol{P},\boldsymbol{V}_1,\boldsymbol{V}_2} \ & \frac{1}{2} \sum_{j=1}^{m} \left\| \left(\mathbf{1}_n - \boldsymbol{p}_j \right) \odot \boldsymbol{y}_j + \boldsymbol{p}_j \odot \boldsymbol{y}_j \odot \boldsymbol{x}_j - \boldsymbol{x}_j \right\|_2^2 \\ & + \lambda \mathcal{R}_1(\boldsymbol{V}_1) + \tau \mathcal{R}_2(\boldsymbol{V}_2) + l_{\mathbb{R}_+^{n\times r}}(\boldsymbol{E}) \\ & + l_{\{\mathbf{1}_m^\top\}}\left(\mathbf{1}_r^\top \boldsymbol{A} \right) + l_{\mathbb{R}_+^{r\times m}}(\boldsymbol{A}) + l_{\{\boldsymbol{P}<\mathbf{1}_{n\times m}\}}(\boldsymbol{P}) \\ \text{s.t.} \ & \boldsymbol{y}_j = \boldsymbol{E}\boldsymbol{a}_j, \quad \boldsymbol{V}_1 = \boldsymbol{A}, \ \boldsymbol{V}_2 = \boldsymbol{P} \end{aligned} \tag{12}$$

where $\iota_{\mathbb{R}_+^{n\times r}}(\boldsymbol{E})$, $\iota_{\{\mathbf{1}_m^\top\}}(\mathbf{1}_r^\top A)$, $\iota_{\mathbb{R}_+^{r\times m}}(\boldsymbol{A})$, and $\iota_{\{\boldsymbol{P}<\mathbf{1}_{n\times m}\}}(\boldsymbol{P})$ are the indicator functions that impose the nonnegative constraint on the endmembers, enforce the abundances to meet the ASC and ANC, and bound the nonlinear parameters to be less than 1, respectively. The augmented Lagrangian function is derived as

$$\begin{aligned} & \mathcal{L}(\boldsymbol{E}, \boldsymbol{A}, \boldsymbol{P}, \boldsymbol{V}_1, \boldsymbol{V}_2, \boldsymbol{U}_1, \boldsymbol{U}_2) \\ & = \frac{1}{2} \sum_{j=1}^{m} \left\| \left(\mathbf{1}_n - \boldsymbol{p}_j \right) \odot \left(\boldsymbol{E}\boldsymbol{a}_j \right) + \boldsymbol{p}_j \odot \left(\boldsymbol{E}\boldsymbol{a}_j \right) \odot \boldsymbol{x}_j - \boldsymbol{x}_j \right\|_2^2 \\ & \quad + \lambda \mathcal{R}_1(\boldsymbol{V}_1) + \tau \mathcal{R}_2(\boldsymbol{V}_2) + l_{\mathbb{R}_+^{n\times r}}(\boldsymbol{E}) + l_{\{\mathbf{1}_m^\top\}}\left(\mathbf{1}_r^\top \boldsymbol{A} \right) + l_{\mathbb{R}_+^{r\times m}}(\boldsymbol{A}) \\ & \quad + l_{\{\boldsymbol{P}<\mathbf{1}_{n\times m}\}}(\boldsymbol{P}) + \frac{\mu}{2} \left(\| \boldsymbol{A} - \boldsymbol{V}_1 + \boldsymbol{U}_1 \|_F^2 + \| \boldsymbol{P} - \boldsymbol{V}_2 + \boldsymbol{U}_2 \|_F^2 \right) \end{aligned} \tag{13}$$

where $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$ are scaled dual variables, and μ is a penalty parameter with a small positive value controlling the convergence speed of the algorithm. Alternately minimizing one variable with the others fixed, the variables in (13) are sequentially optimized.

1) Minimization With Respect to $\boldsymbol{E}$: In the $(t+1)$th iteration, discarding the terms irrelevant with $\boldsymbol{E}$, $\boldsymbol{E}^{(t+1)}$ is updated by minimizing the following optimization subproblem

$$\begin{aligned} & \boldsymbol{E}^{(t+1)} \\ & = \arg\min_{\boldsymbol{E}} \ l_{\mathbb{R}_+^{n\times r}}(\boldsymbol{E}) \\ & \quad + \frac{1}{2} \sum_{j=1}^{m} \left\| \left(\mathbf{1}_n - \boldsymbol{p}_j^{(t)} \right) \odot \left(\boldsymbol{E}\boldsymbol{a}_j^{(t)} \right) + \boldsymbol{p}_j^{(t)} \odot \left(\boldsymbol{E}\boldsymbol{a}_j^{(t)} \right) \odot \boldsymbol{x}_j - \boldsymbol{x}_j \right\|_2^2 \end{aligned} \tag{14}$$

and the equivalent form of the above problem is formulated as

$$\boldsymbol{E}^{(t+1)} = \arg\min_{\boldsymbol{E}} \ \frac{1}{2} \sum_{j=1}^{m} \left\| \left(\boldsymbol{E} \odot \tilde{\boldsymbol{A}}_j^{(t)} \right) \mathbf{1}_r - \boldsymbol{x}_j \right\|_2^2 + l_{\mathbb{R}_+^{n\times r}}(\boldsymbol{E}) \tag{15}$$

where $\tilde{\boldsymbol{A}}_j^{(t)} = (\mathbf{1}_n - \boldsymbol{p}_j^{(t)} + \boldsymbol{p}_j^{(t)} \odot \boldsymbol{x}_j)(\boldsymbol{a}_j^{(t)})^\top$. Then, the gradient projection method [17] is used to update $\boldsymbol{E}^{(t+1)}$ row by row as

$$\boldsymbol{e}_{\lambda_i}^{(t+1)} = \boldsymbol{e}_{\lambda_i}^c - \gamma_{\lambda_i}^{(t+1)} \left(\boldsymbol{e}_{\lambda_i}^c \boldsymbol{C}^{(t+1)} - \boldsymbol{b}_{\lambda_i}^{(t+1)} \right) \tag{16}$$

where $\boldsymbol{e}_{\lambda_i}^{(t+1)}$ and $\boldsymbol{e}_{\lambda_i}^c$ denote the ith row of $\boldsymbol{E}^{(t+1)}$ and $\boldsymbol{E}^c$, respectively. $\gamma_{\lambda_i}^{(t+1)} = 1 / ||\boldsymbol{C}^{(t+1)}||_F$ is the step size, and $||\cdot||_F$ denotes the Frobenius norm. $\boldsymbol{b}_{\lambda_i}^{(t+1)}$ is the ith row of $\boldsymbol{B}^{(t+1)}$

$$\boldsymbol{B}^{(t+1)} = \sum_{j=1}^{m} \begin{bmatrix} \left(\boldsymbol{x}_j \odot \tilde{\boldsymbol{a}}_{j,1}^{(t)} \right)^\top \\ \vdots \\ \left(\boldsymbol{x}_j \odot \tilde{\boldsymbol{a}}_{j,r}^{(t)} \right)^\top \end{bmatrix}^\top \in \mathbb{R}^{n\times r} \tag{17}$$

$$\begin{aligned} & \boldsymbol{C}^{(t+1)} \\ & = \sum_{j=1}^{m} \begin{bmatrix} \left(\tilde{a}_{j,1}^i \right)^{(t)} \left(\tilde{a}_{j,1}^i \right)^{(t)} & \cdots & \left(\tilde{a}_{j,1}^i \right)^{(t)} \left(\tilde{a}_{j,r}^i \right)^{(t)} \\ \vdots & & \vdots \\ \left(\tilde{a}_{j,r}^i \right)^{(t)} \left(\tilde{a}_{j,1}^i \right)^{(t)} & \cdots & \left(\tilde{a}_{j,r}^i \right)^{(t)} \left(\tilde{a}_{j,r}^i \right)^{(t)} \end{bmatrix} \in \mathbb{R}^{r\times r} \end{aligned} \tag{18}$$

where $\tilde{\boldsymbol{A}}_j^{(t)} = [\tilde{\boldsymbol{a}}_{j,1}^{(t)}, \cdots \tilde{\boldsymbol{a}}_{j,r}^{(t)}]$, and $(\tilde{a}_{j,r}^i)^{(t)}$ represents the ith element of the vector $\tilde{\boldsymbol{a}}_{j,r}^{(t)} \in \mathbb{R}^n$. Considering the nonnegative constraint on $\boldsymbol{E}^{(t+1)}$, the final solution is written as

$$\boldsymbol{E}^{(t+1)} = \max\left(\boldsymbol{E}^{(t+1)}, 0 \right) \tag{19}$$

2) Minimization With Respect to $\boldsymbol{A}$: In the $(t+1)$th iteration, the optimization of the subproblem with respect to $\boldsymbol{A}$ is given as

$$\begin{aligned} \boldsymbol{A}^{(t+1)} = \arg\min_{\boldsymbol{A}} \ & \frac{\mu}{2} \left\| \boldsymbol{A} - \boldsymbol{V}_1^{(t)} + \boldsymbol{U}_1^{(t)} \right\|_F^2 + l_{\mathbb{R}_+^{r\times m}}(\boldsymbol{A}) + l_{\{\mathbf{1}_m^\top\}}\left(\mathbf{1}_r^\top \boldsymbol{A} \right) \\ & + \frac{1}{2} \sum_{j=1}^{m} \left\| \left(\mathbf{1}_n - \boldsymbol{p}_j^{(t)} \right) \odot \left(\boldsymbol{E}^{(t+1)} \boldsymbol{a}_j \right) \right. \\ & \left. + \boldsymbol{p}_j^{(t)} \odot \left(\boldsymbol{E}^{(t+1)} \boldsymbol{a}_j \right) \odot \boldsymbol{x}_j - \boldsymbol{x}_j \right\|_2^2. \end{aligned} \tag{20}$$

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The solution of (20) can be derived pixelwisely as

$$\boldsymbol{a}_j^{(t+1)} = \left(\left(\tilde{\boldsymbol{E}}_j^{(t+1)}\right)^\top \tilde{\boldsymbol{E}}_j^{(t+1)} + \mu \boldsymbol{I}_{r\times r}\right)^{-1} \times \left(\left(\tilde{\boldsymbol{E}}_j^{(t+1)}\right)^\top \boldsymbol{x}_j + \mu\left(\boldsymbol{v}_{1,j}^{(t)} - \boldsymbol{u}_{1,j}^{(t)}\right)\right) \tag{21}$$

where $\tilde{\boldsymbol{E}}_j^{(t+1)} = \boldsymbol{E}^{(t+1)} \odot ((\boldsymbol{1}_n - \boldsymbol{P}_j^{(t)} + \boldsymbol{P}_j^{(t)} \odot \boldsymbol{x}_j)\boldsymbol{1}_r^\top)$, $\boldsymbol{I}_{r\times r}$ is an identity matrix, and $\boldsymbol{v}_{1,j}^{(t)}$ and $\boldsymbol{u}_{1,j}^{(t)}$ denote the jth column of $\boldsymbol{V}_1$ and $\boldsymbol{U}_1$, respectively. Then, the ANC and ASC are imposed on $\boldsymbol{A}^{(t+1)}$

$$\boldsymbol{A}^{(t+1)} = \max\left(\boldsymbol{A}^{(t+1)}, 0\right), \quad \boldsymbol{A}^{(t+1)} = \boldsymbol{A}^{(t+1)} \cdot / \left(\boldsymbol{1}_{r\times r}\boldsymbol{A}^{(t+1)}\right). \tag{22}$$

3) Minimization With Respect to $\boldsymbol{P}$: The subproblem with respect to $\boldsymbol{P}$ is written as

$$\boldsymbol{P}^{(t+1)} = \arg\min_{\boldsymbol{P}} \{\boldsymbol{P} < \boldsymbol{1}_{n\times m}\}(\boldsymbol{P}) + \frac{\mu}{2}\left\|\boldsymbol{P} - \boldsymbol{V}_2^{(t)} + \boldsymbol{U}_2^{(t)}\right\|_F^2 + \frac{1}{2}\sum_{j=1}^{m}\left\|(\boldsymbol{1}_n - \boldsymbol{p}_j) \odot \left(\boldsymbol{E}^{(t+1)}\boldsymbol{a}_j^{(t+1)}\right) + \boldsymbol{p}_j \odot \left(\boldsymbol{E}^{(t+1)}\boldsymbol{a}_j^{(t+1)}\right) \odot \boldsymbol{x}_j - \boldsymbol{x}_j\right\|_2^2 \tag{23}$$

and the solution is given as

$$\boldsymbol{P}^{(t+1)} = \frac{\mu\left(\boldsymbol{V}_2^{(t)} - \boldsymbol{U}_2^{(t)}\right) + \left(\boldsymbol{Y}^{(t+1)} - \boldsymbol{Y}^{(t+1)} \odot \boldsymbol{X}\right) \odot \left(\boldsymbol{Y}^{(t+1)} - \boldsymbol{X}\right)}{\left(\boldsymbol{Y}^{(t+1)} - \boldsymbol{Y}^{(t+1)} \odot \boldsymbol{X}\right)^2 + \mu \boldsymbol{1}_{n\times m}} \tag{24}$$

where $\boldsymbol{Y}^{(t+1)} = \boldsymbol{E}^{(t+1)}\boldsymbol{A}^{(t+1)}$. Moreover, the boundary constraint should be considered, and $\boldsymbol{P}^{(t+1)}$ is updated as follows:

$$\boldsymbol{P}^{(t+1)} = \min\left(\boldsymbol{P}^{(t+1)}, 1\right) \tag{25}$$

4) Minimization With Respect to $\boldsymbol{V}_1$ and $\boldsymbol{V}_2$: $\boldsymbol{V}_1^{(t+1)}$ and $\boldsymbol{V}_2^{(t+1)}$ are updated in a superpixel-by-superpixel manner, for $q = 1, \ldots, Q$

$$\boldsymbol{V}_{1,q}^{(t+1)} = \arg\min_{\boldsymbol{V}_{1,q}} \lambda \sum_{q=1}^{Q}\sum_{\boldsymbol{v}_{1,j}\in\boldsymbol{V}_{1,q}} c_j \left\|\boldsymbol{W}_q^A \boldsymbol{v}_{1,j}\right\|_1 + \frac{\mu}{2}\left\|\boldsymbol{A}_q^{(t+1)} - \boldsymbol{V}_{1,q} + \boldsymbol{U}_{1,q}^{(t)}\right\|_F^2 \tag{26}$$

where $\boldsymbol{W}_q^A$ is calculated based on (7), and $\boldsymbol{V}_{1,q}, \boldsymbol{U}_{1,q} \in \mathbb{R}^{r\times N_q}$ are the submatrices of $\boldsymbol{V}_1$ and $\boldsymbol{U}_1$ in the qth superpixel. The pixelwise soft threshold solution of (26) is given as

$$\boldsymbol{v}_{1,j}^{(t+1)} = \text{soft}\left(\boldsymbol{a}_j^{(t+1)} + \boldsymbol{u}_{1,j}^{(t)}, \left(c_j\lambda/\mu\right)\boldsymbol{W}_q^A\right) \tag{27}$$

where $\text{soft}(\boldsymbol{x}, \boldsymbol{\theta}) = \text{sign}(\boldsymbol{x})\max\{|\boldsymbol{x}| - \boldsymbol{\theta}, 0\}$ is the *soft threshold* function [44], and $\boldsymbol{a}_j \in \boldsymbol{A}_q$ and $\boldsymbol{u}_{1,j} \in \boldsymbol{U}_{1,q}$

$$\boldsymbol{V}_{2,q}^{(t+1)} = \arg\min_{\boldsymbol{V}_{2,q}} \frac{\tau}{2}\sum_{q=1}^{Q}\sum_{\boldsymbol{v}_{2,j}\in\boldsymbol{V}_{2,q}} c_j \boldsymbol{v}_{2,j}^\top \boldsymbol{L}_q \boldsymbol{v}_{2,j} + \frac{\mu}{2}\left\|\boldsymbol{P}_q^{(t+1)} - \boldsymbol{V}_{2,q} + \boldsymbol{U}_{2,q}^{(t)}\right\|_F^2 \tag{28}$$

Algorithm 2 Pseudocode of CTF-UNHU

Input: Hyperspectral data $\boldsymbol{X} \in \mathbb{R}^{n\times m}$, and parameters κ and κ_s.
Output: Endmember matrix $\boldsymbol{E} \in \mathbb{R}^{n\times r}$, abundance matrix $\boldsymbol{A} \in \mathbb{R}^{r\times m}$, and nonlinear parameter matrix $\boldsymbol{P} \in \mathbb{R}^{n\times m}$.

Pre-processing:
1: Perform the superpixel segmentation on $\boldsymbol{X}$ with the SLIC.
2: Generate the coarse HSIs $\boldsymbol{X}^c$ by averaging the pixels' spectra at each superpixel.
3: Calculate the pixel-wise coefficient c_j based on (7).
Set $t = 0$, $\mu > 0$, $\lambda > 0$, and $\tau > 0$.
Initialization: $\boldsymbol{E}^c$, $\boldsymbol{A}^c$, $\boldsymbol{P}^c$, $\boldsymbol{A}^{(0)}$, $\boldsymbol{P}^{(0)}$, and $\boldsymbol{V}_n^{(0)}$, $\boldsymbol{U}_n^{(0)}$, $n = 1, 2$
While stopping conditions are not met, **do**
  Update $\boldsymbol{E}^c$, $\boldsymbol{A}^c$, and $\boldsymbol{P}^c$ by coarse unmixing.
  Update $\boldsymbol{W}_q^A$ and $\boldsymbol{W}_q^P$ with (6) and (9).
  Update $\boldsymbol{E}^{(t+1)}$, $\boldsymbol{A}^{(t+1)}$, and $\boldsymbol{P}^{(t+1)}$ with (19), (22), and (25), respectively.
  Update $\boldsymbol{V}_1^{(t+1)}$ and $\boldsymbol{V}_2^{(t+1)}$ with (27) and (29), respectively.
  Update $\boldsymbol{U}_1^{(t+1)}$ and $\boldsymbol{U}_2^{(t+1)}$ with (30) and (31), respectively.
  $t = t + 1$.
End

where $\boldsymbol{L}_q$ is calculated based on (10), and $\boldsymbol{V}_{2,q}, \boldsymbol{U}_{2,q} \in \mathbb{R}^{n\times N_q}$ are the submatrices of $\boldsymbol{V}_2$ and $\boldsymbol{U}_2$ in the qth superpixel. The pixelwise solution of (28) is given as

$$\boldsymbol{v}_{2,j}^{(t+1)} = \left(c_j\tau\boldsymbol{L}_q + \mu\boldsymbol{I}_{n\times n}\right)^{-1}\left(\mu\left(\boldsymbol{p}_j^{(t+1)} + \boldsymbol{u}_{2,j}^{(t)}\right)\right) \tag{29}$$

where $\boldsymbol{p}_j \in \boldsymbol{P}_q$ and $\boldsymbol{u}_{2,j} \in \boldsymbol{U}_{1,q}$.

5) Minimization With Respect to $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$: At last, the scaled dual variables $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$ are updated as

$$\boldsymbol{U}_1^{(t+1)} = \boldsymbol{U}_1^{(t)} + \boldsymbol{A}^{(t+1)} - \boldsymbol{V}_1^{(t+1)} \tag{30}$$

$$\boldsymbol{U}_2^{(t+1)} = \boldsymbol{U}_2^{(t)} + \boldsymbol{P}^{(t+1)} - \boldsymbol{V}_2^{(t+1)}. \tag{31}$$

After alternately updating all the variables in the $(t+1)$th iteration, if the algorithm does not satisfy specific stopping conditions, the endmembers $\boldsymbol{E}^{(t+1)}$, and the superpixel-based weighted average of abundances and nonlinear parameters, i.e., $\sum_{j=1}^{N_q} c_j \boldsymbol{a}_j^{(t+1)} / \sum_{h=1}^{N_q} c_h$ and $\sum_{j=1}^{N_q} c_j \boldsymbol{p}_j^{(t+1)} / \sum_{h=1}^{N_q} c_h$ $(q = 1, \ldots, Q)$, are fed back to initialize the coarse unmixing. The pseudocode of CTF-UNHU is provided in Algorithm 2.

D. Implementation Details

In the coarse unmixing stage, the endmembers extracted from the coarse HSIs via the VCA, the coarse abundances estimated by the FCLS, and the coarse nonlinear parameters calculated based on (24) are taken as the first initialization of $\boldsymbol{E}^c$, $\boldsymbol{A}^c$, and $\boldsymbol{P}^c$, respectively. As stated in [27], the extracted region-based endmembers can provide a more accurate initialization. With the initial endmembers $\boldsymbol{E}^c$, $\boldsymbol{A}^{(0)}$ and $\boldsymbol{P}^{(0)}$ are estimated in the same way as in the coarse unmixing stage. Let $\boldsymbol{V}_1^{(0)}$ and $\boldsymbol{V}_2^{(0)}$ be equal to $\boldsymbol{A}^{(0)}$ and $\boldsymbol{P}^{(0)}$, and $\boldsymbol{U}_1^{(0)}$ and $\boldsymbol{U}_2^{(0)}$ be the zero matrices with different sizes.

In both coarse and fine unmixing stages, two stopping conditions are considered: 1) the number of iterations should not exceed the corresponding maximum number of iterations and 2) the relative changes of the objective function value are below a predefined threshold.

For the computational complexity of CTF-UNHU, in every iteration, the steps of updating $\boldsymbol{E}$, $\boldsymbol{A}$, and $\boldsymbol{V}_2$ consume more

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time than updating the other variables, and their complexities are $\mathcal{O}(mnr)$, $\mathcal{O}(mn^2r)$, and $\mathcal{O}(mn^3)$, respectively.

IV. Experimental Results

A. Compared Methods and Performance Metrics

Seven state-of-the-art unmixing algorithms are selected for comparative experiments, including two LMM-based algorithms (FCLS [4] and LNMF [6]), two BMM-based algorithms (PPNM [13] and BCNMF [14]), two MLM-based algorithms (MLM [16] and MLMp [17]), and one EMLM-based algorithm (BNLSU [23]). Among them, PPNM, MLM, and BNLSU only serve for abundance estimation, and the others jointly estimate the abundances and endmembers. All parameters involved in all the compared methods are empirically tuned or set as recommended by the original literatures. The algorithms are executed using MATLAB R2020b on a computer with an Intel Core i9-9900K CPU at 3.60 GHz and 32-GB RAM.

The performances of the abundance estimation and endmember extraction are evaluated with the root mean square error (RMSE) and the mean spectral angle distance (MSAD)

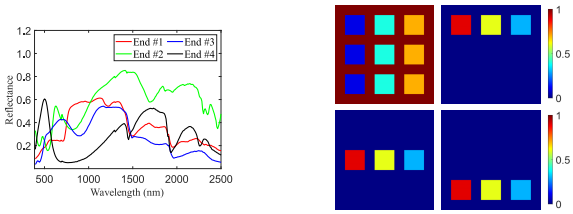
$$\begin{aligned}\text{RMSE} &= \sqrt{\frac{1}{rm}\sum_{j=1}^{m}\left\|\boldsymbol{a}_j-\hat{\boldsymbol{a}}_j\right\|_2^2}\\ \text{MSAD} &= \frac{1}{r}\sum_{i=1}^{r}\cos^{-1}\left(\frac{\boldsymbol{e}_i^{\top}\hat{\boldsymbol{e}}_i}{\|\boldsymbol{e}_i\|_2\left\|\hat{\boldsymbol{e}}_i\right\|_2}\right)\end{aligned} \tag{32}$$

where $\hat{\boldsymbol{a}}_j$ and $\hat{\boldsymbol{e}}_i$ represent the estimated abundance vector of the jth pixel and the ith endmember spectrum, respectively. Since RMSE and MSAD measure the similarity of the estimated variables and the ground truth, the smaller their values, the better the unmixing accuracy.

Besides, the estimated nonlinear intensity can be interpreted by removing the estimated linear component from the data reconstruction, i.e., $\hat{\boldsymbol{X}}_{\text{Non}}=\hat{\boldsymbol{X}}-\hat{\boldsymbol{X}}_{\text{Lin}}\in\mathbb{R}^{n\times m}$ [24]. Instead of performing numerical comparison, $\hat{\boldsymbol{X}}_{\text{Non}}$ estimated by different methods is compared by the spatial and spectral dimensions, i.e., $\boldsymbol{NI}_p\in\mathbb{R}^m$ and $\boldsymbol{NI}_b\in\mathbb{R}^n$ as follows:

$$\begin{aligned}\boldsymbol{NI}_p &= \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left\|\left(\hat{\boldsymbol{X}}_{\text{Non}}\right)_{i,:}\right\|_2^2}\\ \boldsymbol{NI}_b &= \sqrt{\frac{1}{m}\sum_{j=1}^{m}\left\|\left(\hat{\boldsymbol{X}}_{\text{Non}}\right)_{:,j}\right\|_2^2}.\end{aligned} \tag{33}$$

By folding the vector $\boldsymbol{NI}_p$ into a 2-D matrix with spatial coordinates, the spatial distribution map of nonlinear intensity can be drawn. The curve related to the vector $\boldsymbol{NI}_b$ is plotted to show the nonlinear intensity at different bands. Since the true nonlinear intensity is only available for experiments on the synthetic data, the estimated nonlinear intensity can be compared from both the spatial and spectral dimensions. Although the true nonlinear intensity is unavailable for experiments on real datasets, they are qualitatively compared according to the estimated nonlinear intensity due to that the strong nonlinear mixing effects are possibly observed at the junctions of different materials in real scenes.



(a) (b)

Fig. 2. Ground truth of the synthetic data. (a) Four endmember spectra were selected from the USGS spectral library. (b) True abundance maps.

B. Experiments on the Synthetic Data

1) Data Description: The synthetic data consist of 50 × 50 pixels generated using four endmembers selected from the U.S. Geological Survey (USGS) spectral library [45]. As shown in Fig. 2(a), the selected endmember spectra have 224 bands covering a spectral range of wavelengths from 380 to 2500 nm. In Fig. 2(b), the true abundance maps are provided. There are nine squares arranged in a 3 × 3-pixel pattern. Each square, with a size of 5 × 5 pixels, has the same abundances. Background pixels contain only the first endmember, denoted as End #1. For the top to the bottom row of squares, the abundances of End # 2, #3, and #4 vary from 0.3 to 0.9 with an interval of 0.3 in sequence. To satisfy the ASC, in different squares, the complement values of the abundances of End # 2, #3, and #4 were taken as the abundances of End #1 [46].

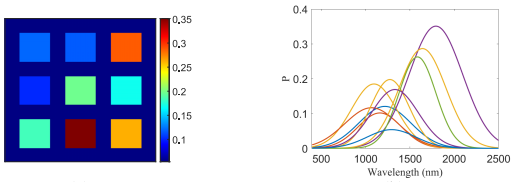
To generate the EMLM-based synthetic data, the nonlinear parameters $\boldsymbol{P}$ should be rationally constructed. According to the works in [23] and [24], it is assumed that the distribution of $\boldsymbol{P}_j$ $(j=1,\ldots,m)$ is akin to the shape of the single Gaussian function, and the maximum values of the bandwise nonlinear parameters, denoted as $\boldsymbol{P}_{\max}\in\mathbb{R}^{1\times m}$, have spatial correlations. Generated by drawing samples from a half-normal distribution with $\sigma=0.3$, the values of $\boldsymbol{P}_{\max}$ keep unchanged in the background and the same square, where the smallest is assigned to the background with low nonlinearity and the rest are randomly assigned to the nine squares, as shown in Fig. 3(a). For the jth pixel, the band index BI_j of $P_{\max,j}$ is randomly selected from 70 to 150, and $P_{i,j}$ at the other bands can be calculated as

$$P_{i,j}=P_{\max,j}\exp\left(-\left(i-BI_j\right)^2/2\sigma_j^2\right) \tag{34}$$

where σ_j is uniformly drawn from (20, 35), regulating the amplitude of $\boldsymbol{P}_j$'s distribution. With the endmembers, abundances, and nonlinear parameters in Fig. 3(b), the mixed pixels can be generated based on the EMLM model. To test the algorithm's robustness to noise, the additive white Gaussian noise is added to the data, where the noise levels, denoted by the signal-to-noise ratio (SNR), vary from 10 to 30 dB.

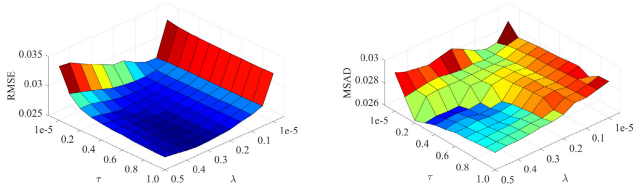
2) Parameter Analysis: In CTF-UNHU, the maximum numbers of iterations were set to 100 for both the outer loop (i.e., fine unmixing) and inner loop (i.e., coarse unmixing), and the predefined thresholds were accordingly set to 10^{-4} and 10^{-6}. Taking the synthetic data with SNR = 30 dB as a studying case, we first analyze the impact of the regularizer parameters on the unmixing accuracy, where the

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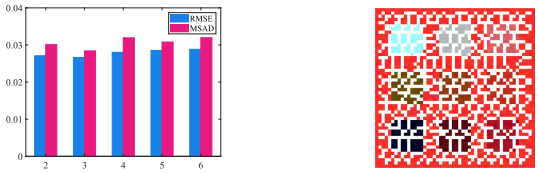
(a) (b)

Fig. 3. Construction of the nonlinear parameters $\boldsymbol{P}$. (a) Spatial distribution of $\boldsymbol{P}_{\max}$. (b) Single Gaussian distribution of $\boldsymbol{P}_j$ $(j = 1, \ldots, m)$ at each band.



(a) (b)

Fig. 4. Regularizer parameter analysis of CTF-UNHU. With different combinations of regularizer parameters λ and τ, variations in (a) RMSE and (b) MSAD.



(a) (b)

Fig. 5. Superpixel parameter analysis of CTF-UNHU. (a) With different κ, variations in RMSE and MSAD. (b) Generated superpixels when $\kappa = 3$, where the RGB color image is composited by bands 56, 112, and 168.

parameters λ and τ were, respectively, selected in the range of $(10^{-5}, 0.5)$ and $(10^{-5}, 1)$. The RMSE and MSAD values obtained by CTF-UNHU with different parameter combinations are depicted in Fig. 4(a) and (b). It can be observed that the changes in terms of both RMSE and MSAD are gentle, and an appropriate solution can be reached when $\lambda = 0.3$ and $\tau = 0.5$. Note that the parameter settings were kept fixed throughout the experiments.

Next, the influence of superpixel parameters on the unmixing accuracy is analyzed. There are two parameters, i.e., κ_s and κ, in the SLIC. κ_s weights the relative importance between the spectral and spatial similarities, and κ controls the average size of the superpixels. Following the work in [38], κ_s was set to 0.3 provided that spectral proximity contributes more to measuring similarity. κ had a value in the range of (2, 6), and the impact of its changes on both RMSE and MSAD is shown in Fig. 5(a). It can be seen that the unmixing accuracy of our method is insensitive to changes in κ over a relatively wide range. Therefore, κ was set to 3 in the following experiments. In Fig. 5(b), the generated superpixels are shown when $\kappa = 3$.

3) Ablation Experiments: To explicitly analyze the contributions of the EMLM model, the number of regularizers, and the coarse-to-fine scheme to the unmixing accuracy of our method, the related ablation experiments are conducted. As listed in Table I, the following cases with different configurations are considered: 1) MUNHU and EUNHU are two methods for the MLM/EMLM-based unsupervised nonlinear hyperspectral unmixing, abbreviated as MU and EU; 2) MUR (MU with a regularizer on $\boldsymbol{A}$), EUR (EU with a regularizer on $\boldsymbol{A}$), and EUR^2 (EU with two regularizers on $\boldsymbol{A}$ and $\boldsymbol{P}$), where the weight matrices in these regularizers are constructed globally without considering the spatial correlations between neighboring pixels; and 3) CTF-MUR, CTF-EUR, and CTF-UNHU combine the coarse-to-fine scheme with MUR, EUR, and EUR^2. The above methods are optimized based on the AOS and ADMM, and the parameters involved in them were tuned to achieve optimal accuracy. Accordingly, the parameter settings are provided in Table I.

To compare the unmixing accuracy of the methods listed in Table I, we run them 20 times on the synthetic data in the case of SNR = 30 dB with randomly distributed noise. In Table II, the mean values of RMSE and MSAD with the standard deviations (std) are provided, where the best scores are bolded. By observing the experimental results, the following conclusions can be drawn.

First, with the same configuration, the unmixing accuracy obtained by the EMLM-based methods is superior to that obtained by the MLM-based methods. Second, incorporating more regularizers in the objective function leads to improved unmixing accuracy, and on this basis, the gain of unmixing accuracy can be further increased by using the coarse-to-fine scheme. Third, although the application of the EMLM model, the regularizers on both $\boldsymbol{A}$ and $\boldsymbol{P}$, and the coarse-to-fine scheme can separately improve the unmixing accuracy to a certain extent, integrating them in the proposed method can lead to the largest gain in the unmixing accuracy.

TABLE I

METHODS WITH DIFFERENT CONFIGURATIONS FOR ABLATION EXPERIMENTS

w/o regularizers	MU	EU	
w/o CTF	MUR (0.5)	w/o $\mathcal{R}(\boldsymbol{P})$	w/ $\mathcal{R}(\boldsymbol{P})$
		EUR (0.1)	EUR^2 (0.1, 0.5)
w/ CTF	CTF-MUR (0.5)	CTF-EUR (0.05)	CTF-UNHU (0.3, 0.5)

TABLE II

RMSE AND MSAD VALUES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS FOR ABLATION EXPERIMENTS OVER 20 RUNS ON THE SYNTHETIC DATA WITH SNR = 30 dB

Metrics	MU	MUR	CTF-MUR	EU	EUR	CTF-EUR	EUR^2	CTF-UNHU
RMSE	0.0406 ±0.0007	0.0330 ±0.0027	0.0296 ±0.0001	0.0407 ±0.0011	0.0293 ±0.0002	0.0281 ±0.0005	0.0282 ±0.0002	**0.0266 ±0.0001**
MSAD	0.0357 ±0.0002	0.0328 ±0.0006	0.0331 ±0.0004	0.0320 ±0.0004	0.0296 ±0.0004	0.0286 ±0.0005	0.0309 ±0.0004	**0.0278 ±0.0005**

TABLE III

RMSE VALUES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS FOR COMPARATIVE EXPERIMENTS OVER 20 RUNS ON THE SYNTHETIC DATA

SNR	V/F	LNMF	PPNM	BCNMF	MLM	MLMp	BNLSU	CTF-UNHU
10 dB	0.0949 ±0.0164	0.0794 ±0.0048	0.0976 ±0.0156	0.1038 ±0.0142	0.0932 ±0.0159	0.1057 ±0.0084	0.1584 ±0.0077	**0.0631 ±0.0031**
20 dB	0.0527 ±0.0059	0.0514 ±0.0021	0.0476 ±0.0067	0.0486 ±0.0065	0.0453 ±0.0066	0.0699 ±0.0027	0.0806 ±0.0020	**0.0271 ±0.0002**
30 dB	0.0458 ±0.0010	0.0519 ±0.0009	0.0346 ±0.0010	0.0334 ±0.0011	0.0338 ±0.0010	0.0406 ±0.0007	0.0493 ±0.0011	**0.0266 ±0.0001**

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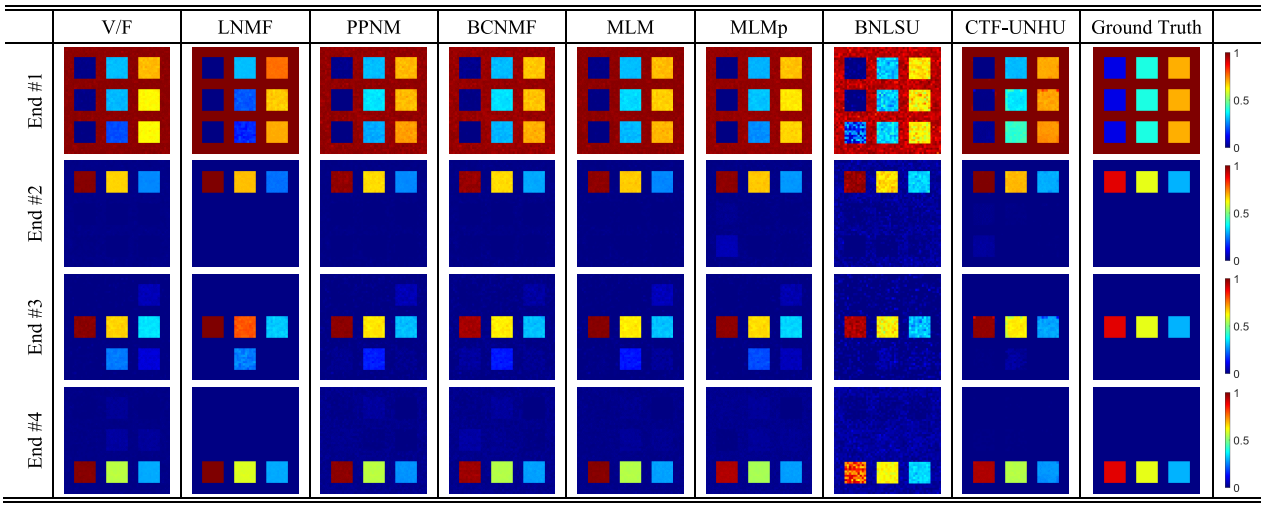


Fig. 6. Abundance maps generated by different methods for the synthetic data with SNR = 30 dB.

TABLE IV
MSAD VALUES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS FOR COMPARATIVE EXPERIMENTS OVER 20 RUNS ON THE SYNTHETIC DATA

SNR	V/F	LNMF	BCNMF	MLMp	CTF-UNHU
10 dB	0.0527 ±0.0035	0.0609 ±0.0014	0.0533 ±0.0033	0.3225 ±0.0145	**0.0504 ±0.0029**
20 dB	0.0322 ±0.0007	0.0420 ±0.0007	0.0322 ±0.0008	0.1678 ±0.0026	**0.0292 ±0.0005**
30 dB	0.0331 ±0.0003	0.0413 ±0.0003	0.0329 ±0.0004	0.0357 ±0.0002	**0.0278 ±0.0005**

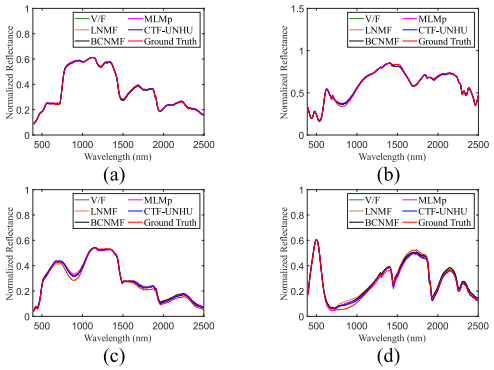


Fig. 7. Endmember spectra extracted by different methods for the synthetic data with SNR = 30 dB. (a) End #1. (b) End #2. (c) End #3. (d) End #4.

4) Comparison With Other Unmixing Methods: To evaluate the unmixing accuracy and the robustness to noise of different methods, we run the methods 20 times on the synthetic data. It can be seen from Table III that the unmixing accuracy deteriorates with the decrease of SNR. Unable to account for the nonlinear mixing effects, the LMM-based methods yield limited RMSEs, where V/F is the abbreviation of VCA-FCLS, and this notation is used in the following experiments. The BMM-based methods obtain good RMSE values, while they are slightly inferior to that of the MLM. MLMp and BNLSU obtain unsatisfactory results that are not only poor in the unmixing accuracy but also sensitive to noise. MLMp implements the difficult unsupervised nonlinear unmixing problem, where the wrongly estimated endmembers can affect the abundance inversion. BNLSU suffers from overfitting, and its unmixing accuracy is greatly influenced by noise, especially in the low SNR case. CTF-UNHU yields the best scores in all cases, indicating that it achieves high-accuracy unmixing and is insensitive to noise. In Table IV, we compare the endmembers extracted by the VCA and four unsupervised

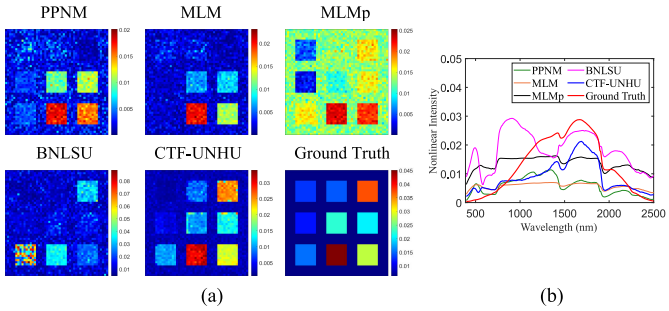


Fig. 8. Nonlinear intensity estimated by different methods for the synthetic data with SNR = 30 dB. (a) Nonlinear intensity at each pixel. (b) Nonlinear intensity at each band.

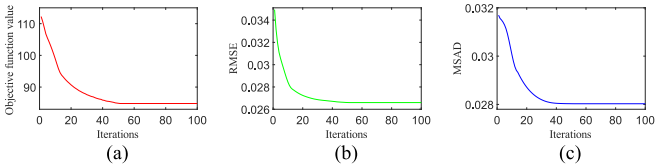


Fig. 9. Convergence analysis of CTF-UNHU. (a) Objective function values, (b) RMSE values, and (c) MSAD values, over 100 iterations.

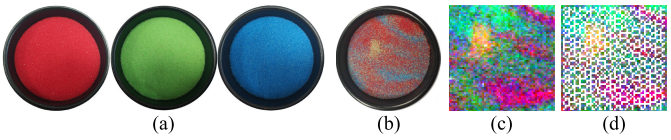


Fig. 10. Laboratory-created data. (a) Different colors of quartz sands with a diameter of 0.3 mm, which serve as pure materials for providing endmembers. (b) Mixture of sand with a spatial pattern. (c) Pseudocolor image, where the RGB color image is composited by bands 64, 128, and 192. (d) Generated superpixels.

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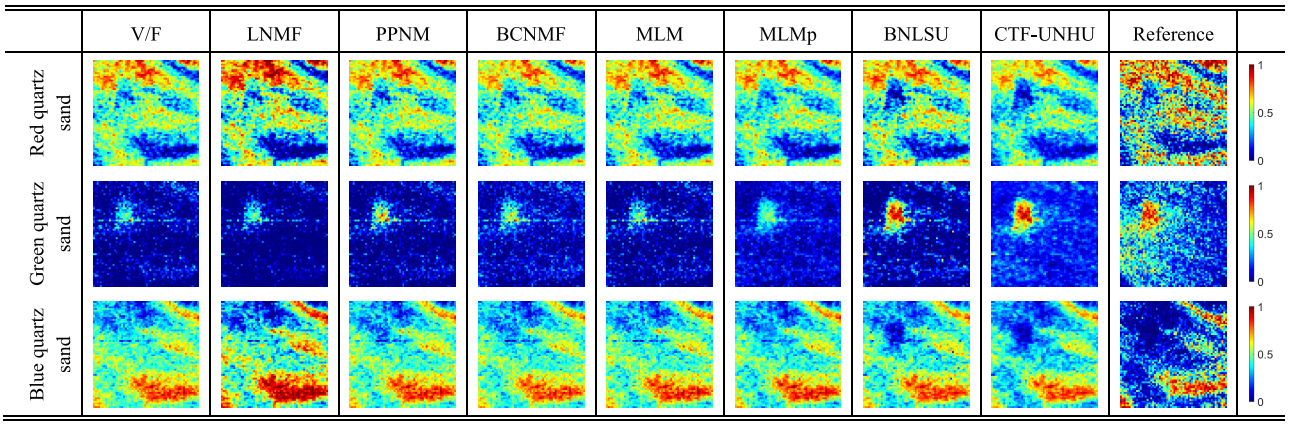


Fig. 11. Abundance maps generated by different methods for the laboratory-created data.

TABLE V

RMSE VALUES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS IN A SUPERVISED MANNER FOR COMPARATIVE EXPERIMENTS OVER 20 RUNS ON THE SYNTHETIC DATA

SNR	FCLS	LNMF(s)	PPNM	BCNMF(s)	MLM	MLMp(s)	BNLSU	CTF-UNHU(s)
10 dB	0.0597 ±0.0008	0.0568 ±0.0008	0.0531 ±0.0006	0.0546 ±0.0006	0.0494 ±0.0006	0.1082 ±0.0016	0.1456 ±0.0055	**0.0480 ±0.0005**
20 dB	0.0426 ±0.0003	0.0408 ±0.0003	0.0257 ±0.0003	0.0237 ±0.0003	0.0251 ±0.0002	0.0329 ±0.0003	0.0736 ±0.0008	**0.0203 ±0.0003**
30 dB	0.0399 ±0.0001	0.0382 ±0.0001	0.0219 ±0.0001	0.0187 ±0.0001	0.0208 ±0.0001	0.0242 ±0.0002	0.0397 ±0.0003	**0.0180 ±0.0001**

unmixing methods. The proposed method achieves the best MSAD values in different cases, while the other methods are slightly inferior in the endmember extraction.

Fig. 6 shows the abundance maps generated by all the compared methods. The abundance maps generated by the proposed method are the most similar to the ground truth. The abundances of pixels in both the background and squares are accurately estimated. The extracted endmember spectra of different methods are provided in Fig. 7, and the proposed method better approximates the ground truth. As shown in Fig. 8, the nonlinear intensity maps and curves generated by using (33) are compared. Since the LMM-based methods fail to model the nonlinearity, they were not considered in this comparison. The proposed method, by mathematically modeling the correlations of the nonlinear parameters along the spatial and spectral dimension, can better reflect the nonlinear intensity at different spatial locations and spectral bands. As a result, the proposed method can better estimate the nonlinear intensity in comparison with the other methods.

5) Comparison Experiments in a Supervised Manner: The above methods all assume that the endmembers are unknown, and the endmembers extracted by the VCA are either fixed in the supervised methods or used as the initialization of the unsupervised methods. In this experiment, only the abundance estimation performance of different methods is compared, supposing that the true endmembers are known. By simply fixing the endmembers, the unsupervised methods in the experiments can be performed in a supervised manner, denoted by the LNMF(s), BCNMF(s), MLMp(s), and CTF-UNHU(s).

As shown in Table V, the unmixing accuracy of all methods gets improved with the real endmembers fixed in unmixing. The LMM-based methods appear to be less competitive

TABLE VI

EXECUTION TIME (IN SECONDS) OF DIFFERENT METHODS IN ABLATION/COMPARISON EXPERIMENTS

Ablation experiments	Methods	MU	MUR	CTF-MUR	EU	EUR	CTF-EUR	EUR^2	CTF-UNHU
	Time (s)	**1.39**	6.19	9.43	2.21	11.63	45.65	14.53	100.68
Comparison experiments	Methods	V/F	LNMF	PPNM	BCNMF	MLM	MLMp	BNLSU	CTF-UNHU
	Time (s)	**0.44**	2.29	3.28	2.19	5.78	1.46	63.02	100.68

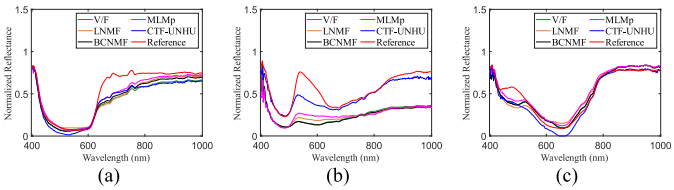


Fig. 12. Endmember spectra extracted by different methods for the laboratory-created data. (a) Red quartz sand. (b) Green quartz sand. (c) Blue quartz sand.

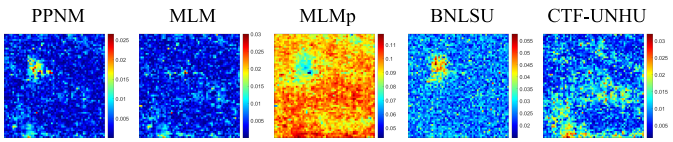


Fig. 13. Nonlinear intensity maps estimated by different methods for the laboratory-created data.

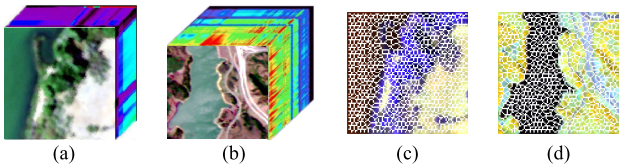


Fig. 14. Real hyperspectral data. Data cube of (a) Samson data and (b) Jasper Ridge data. Generated superpixels for (c) Samson data, where the RGB image is composited by bands 39, 78, and 117, and (d) Jasper Ridge data, where the RGB image is composited by bands 53, 102, and 170.

than those methods considering the nonlinear mixing effects. Except for CTF-UNHU(s), BCNMF(s) slightly outperforms the other methods at relatively high SNR cases. MLM ranks the second in the low SNR case. MLMp(s) and BNLSU cannot get good unmixing accuracy, which can mainly be attributed to the lack of necessary priors and the overfitting of noise. The proposed method yields the best RMSEs at different noise

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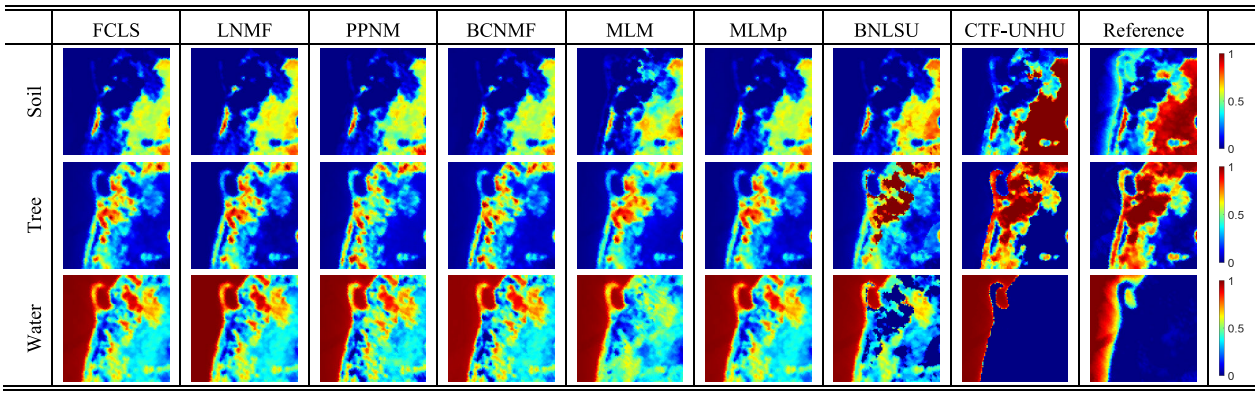


Fig. 15. Abundance maps generated by different methods for Samson data.

levels, showing its advantages in achieving high unmixing accuracy and overcoming overfitting.

6) Convergence and Efficiency Analysis: To illustrate the convergence of the proposed method, the changing curves of the objective function, RMSE, and MSAD values obtained by CTF-UNHU are depicted. As can be observed in Fig. 9, they converge to stable values quickly. In a word, the proposed method has good convergence and can achieve satisfactory unmixing results after sufficient iterations.

For conducting a thorough efficiency analysis, the execution time of different methods is provided in Table VI. In the ablation experiments, using the EMLM model, more regularizers, and the coarse-to-fine scheme accordingly increase the time burden within an acceptable range. In the comparative experiments, the two EMLM-based unmixing methods are more time-consuming than the others, where CTF-UNHU, which includes the endmember extraction, takes a longer running time than that of BNLSU.

C. Experiments on Laboratory-Created Data

1) Data Description: Collected by a push-broom imaging spectrometer, the data consist of 256 spectral bands covering the visible and near-infrared wavelengths ranging from 400 to 1000 nm with a spectral resolution up to 0.58 nm. Different colors of quartz sands with uniform size in Fig. 10(a) were treated as pure materials and then mixed to obtain a mixture of colored quartz sand with a spatial pattern, as shown in Fig. 10(b). Using the spatial resolution ratio, the aligned high-resolution RGB image of this scene can be linked to hyperspectral pixels. The reference abundances can be obtained by analyzing the percentage of each colored sand in the low-resolution hyperspectral pixels with the help of the RGB image. A 60 × 60-pixel subimage cropped from the center of the image was investigated. The pseudocolor image and the generated superpixels are provided in Fig. 10(c) and (d). More details about this laboratory-created data can be found in [47].

2) Experimental Results and Analysis: As shown in Table VII, the linear unmixing methods achieve unsatisfactory accuracy, which indicates that they cannot well address the unmixing problem in this scene with apparent nonlinear mixing effects. Due to the limited accuracy of the endmember extraction via the VCA, the RMSE values obtained by the supervised nonlinear unmixing methods are not as good as their corresponding unsupervised ones. In such a situation, BNLSU yields a good score in RMSE, which shows a great potential for considering wavelength-dependent nonlinear intensity in unmixing. The proposed method achieves the state-of-the-art results in terms of both RMSE and MSAD.

TABLE VII
RMSE AND MSAD VALUES OBTAINED BY DIFFERENT METHODS ON THE LABORATORY-CREATED DATA

Metrics	V/F	LNMF	PPNM	BCNMF	MLM	MLMp	BNLSU	CTF-UNHU
RMSE	0.2734	0.2912	0.2701	0.2611	0.2719	0.2549	0.2591	**0.2268**
MSAD	0.1796	0.1770	-	0.1777	-	0.1337	-	**0.1194**

Furthermore, the estimated abundance maps of different methods are shown in Fig. 11. The spatial patterns of the abundance maps generated by the proposed method are more consistent with the reference ones. In Fig. 12, the extracted endmember spectra of the unsupervised unmixing methods are provided. Endmember spectra extracted by the proposed method are the closest to the reference spectra. The nonlinear intensity maps estimated by the nonlinear unmixing methods are compared as well. As shown in Fig. 13, the proposed method can rationally generate the nonlinear intensity map, which is consistent with the rule that stronger nonlinear mixing effects can be observed at the junctions of different materials.

D. Experiments on Real Hyperspectral Data

1) Data Description: Samson: In Fig. 14(a), Samson data have 95 × 95 pixels with 156 channels covering the wavelength from 400 to 900 nm.There are mainly three endmembers, i.e., soil, tree, and water, in this area. The generated superpixels are shown in Fig. 14(b).

Jasper Ridge: In Fig. 14(c), Jasper Ridge data have 100 × 100 pixels covering 224 channels ranging from 380 to 2500 nm. After removing the channels (1–3, 108–112, 154–166, and 220–224) contaminated by dense water vapor and atmospheric effects, 198 channels were finally retained for unmixing. In this area, there are primarily four endmembers, i.e., tree, water, dirt, and road. The generated superpixels are shown in Fig. 14(d).

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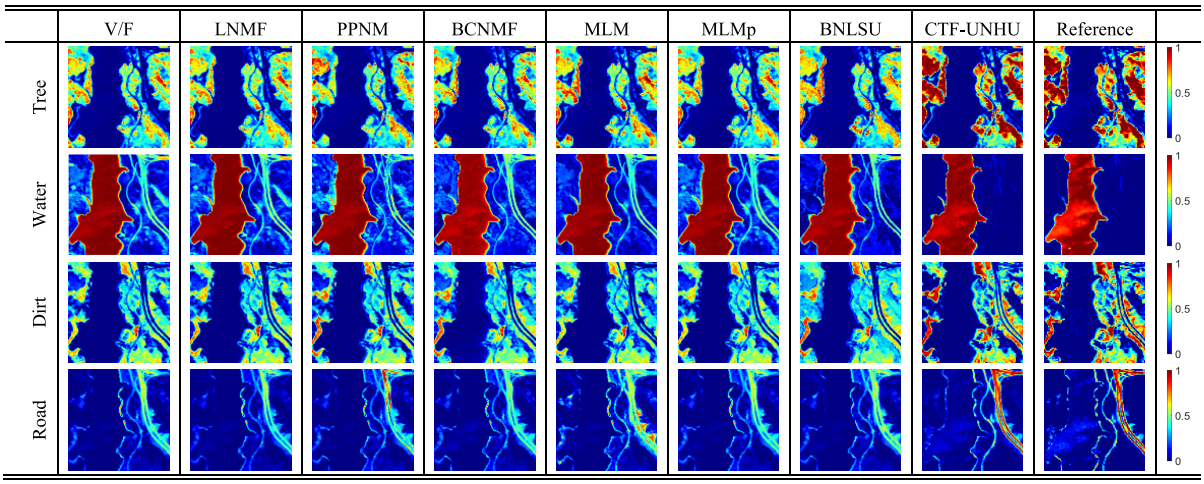


Fig. 16. Abundance maps generated by different methods for Jasper Ridge data.

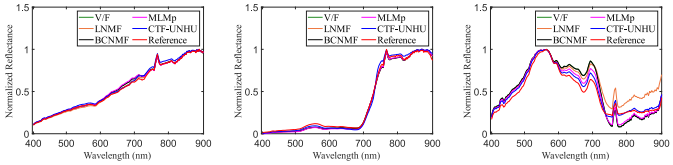


Fig. 17. Endmember spectra extracted by different methods for Samson data. (a) Soil. (b) Tree. (c) Water.

TABLE VIII

RMSE AND MSAD VALUES OBTAINED BY DIFFERENT METHODS ON REAL HYPERSPECTRAL DATA

Data	Metrics	V/F	LNMF	PPNM	BCNMF	MLM	MLMp	BNLSU	CTF-UNHU
Samson	RMSE	0.3243	0.3233	0.3217	0.3228	0.2915	0.3198	0.2605	**0.1365**
	MSAD	0.0713	0.0726	-	0.0714	-	0.0616	-	**0.0343**
Jasper Ridge	RMSE	0.1577	0.1558	0.1551	0.1529	0.1497	0.1571	0.1409	**0.0494**
	MSAD	0.0993	0.0903	-	0.0994	-	0.0958	-	**0.0308**

2) Experimental Results and Analysis: Referring to the ground truth of Samson data and Jasper Ridge data provided in [47], the unmixing methods were evaluated in terms of RMSE and MSAD. As shown in Table VIII, the proposed method achieves high-accuracy estimates of the abundances and endmembers. As shown in Figs. 15 and 16, the abundance maps generated by the proposed method are more distinguishable, compared to that of the other methods, and can accurately reflect the spatial distribution of various endmembers. In Figs. 17 and 18, the extracted endmember spectra of the proposed method appear to be more in line with the reference ones. Moreover, in Fig. 19, the nonlinear intensity map generated by our method shows a more reasonable distribution, and the concentrated nonlinear intensities occur at the material boundaries, which better matches the real situation.

E. Discussion

According to the experiments on the synthetic, laboratory-created, and real hyperspectral data, the advantages of the proposed method can be summarized as follows:

1) *Higher unmixing accuracy:* The proposed method achieves high-accuracy estimates of both the abundances and endmembers by making contributions in the following three aspects: 1) based on the EMLM, the wavelength-dependent nonlinear intensity can be interpreted; 2) by adopting a coarse-to-fine unmixing scheme, the spatial information of HSIs can be well expressed during the unmixing process; and 3) with the help of the coarse unmixing results, a weighted sparse regularizer of abundances and a weighted graph regularizer of nonlinear parameters are constructed and integrated into the objective function, effectively utilizing the priors of at the spectral and spatial dimensions for abundances and nonlinear parameters. By conducting extensive ablation experiments, the impact of the above contributions on the unmixing accuracy is explicitly investigated. It is concluded that the best scores can be achieved by using them together.
2) *Robustness to noise:* In the experiments testing the robustness of different unmixing methods to noise, compared to the state-of-the-art methods, our method can

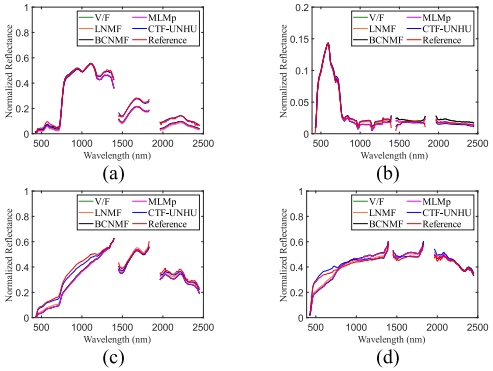


Fig. 18. Endmember spectra extracted by different methods for Jasper Ridge data. (a) Tree. (b) Water. (c) Dirt. (d) Road.

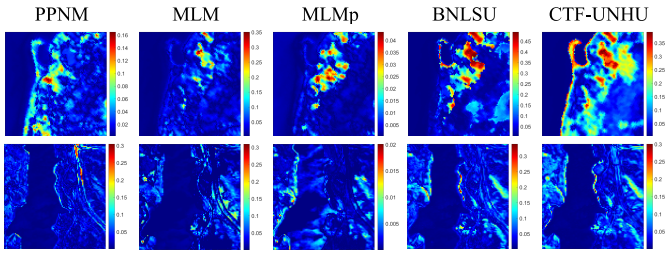


Fig. 19. Nonlinear intensity maps estimated by different methods for Samson data (top) and Jasper Ridge data (bottom).

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provide more accurate estimates of the abundances and endmembers, even when the data are contaminated by severe noise. It may be useful for practical applications due to its low data quality requirements.

3) *Superiority of the Coarse-to-Fine Scheme:* From the experimental results on different types of data, especially the real data, we can observe the superiority of using the coarse-to-fine unmixing scheme. We attribute the merits to the following reasons: on the one hand, the way of generating coarse HSIs by averaging the pixels' spectra in each superpixel effectively enhances the SNR of the candidate endmembers; therefore, our method achieves improved accuracy in the endmember extraction. On the other hand, the coarse abundances and nonlinear parameters are used to guide the regularizer design, so the spatial information of HSIs can be well exploited in unmixing.

In addition, the combination of the EMLM model, weighted regularizers, and the coarse-to-fine scheme enables CTF-UNHU to obtain high unmixing accuracy and enhanced robustness to noise. However, as shown in the efficiency analysis, the proposed method takes much longer time than other methods in ablation/comparative experiments. To alleviate that, some attemptable methods include: 1) GPU accelerated algorithms can be considered; and 2) using the deeply unfolding technology, the procedures of solving the constraint optimization problem can be transformed into a deep learnable neural network to achieve faster convergence and higher unmixing accuracy.

V. Conclusion

In this article, based on the EMLM model, a coarse-to-fine scheme has been proposed for unsupervised nonlinear hyperspectral unmixing. It focuses on solving two major problems, i.e., how to properly establish the unmixing framework and derive the algorithm to effectively solve the unsupervised nonlinear unmixing problem and how to design regularizers to accurately model the priors that well exploit the rich spectral and spatial information in HSIs. First, the coarse HSIs constructed based on the superpixel segmentation were unmixed. Then, with the guidance of the coarse unmixing results, the fine unmixing on the original HSIs was successfully implemented. In the fine unmixing stage, the region-based endmembers were used to update the endmembers, and the coarse abundances and nonlinear parameters were integrated into the regularizer design. Specifically, a weighted sparse regularizer of abundances and a weighted graph regularizer of nonlinear parameters have been constructed to correctly model the priors, i.e., the neighboring pixels share similar sparsity patterns in the abundances and show the consistency in correlations between different bands of the nonlinear parameters. Finally, the two regularizers have been incorporated into the objective function to formulate the optimization problem, and the corresponding algorithm has been derived based on the AOS and ADMM. The experimental results on the synthetic, laboratory-created, and real hyperspectral data have shown that the proposed method can achieve higher accuracy of unmixing results and exhibit robustness to noise, in comparison with the other state-of-the-art unmixing methods.

In future work, a research focus can be the improvement of the efficiency of CTF-UNHU. Inspired by the deep unfolding technology, we intend to derive a deep neural network architecture by unfolding CTF-UNHU to achieve fast convergence and further improved unmixing accuracy.

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